Cantera vs NN | NRMSE summary (util.py: RMS normalization)

Median of per-IC NRMSE on the same 15 complete test ICs for every model

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**10k**	0.0381	0.379	1.38	1.02	0.176	0.967	0.0661	1.02	1.23	0	18/18
2 P-only DeepONet10	**10k**	0.0606	0.124	0.97	0.997	0.177	1	0.157	0.997	1.38	0	18/18
2 P-only DeepONet10	**long**	0.034	0.365	0.878	0.952	0.173	0.324	0.301	0.28	0.482	0	18/18
3 ANN10 + P-only DeepONet10	**10k**	0.0447	0.119	0.906	0.928	0.276	0.998	0.126	0.992	1	0	18/18
3 ANN10 + P-only DeepONet10	**long**	0.0104	0.238	0.854	0.379	0.27	0.199	0.0369	0.321	0.46	0	18/18
4 T/P DeepONet10	**10k**	0.0466	0.138	1.02	0.786	0.519	0.998	0.107	1.15	1.35	0	17/18
4 T/P DeepONet10	**long**	0.307	0.221	6.11	0.698	0.283	3.32	0.259	1.01	1.01	0	16/18
5 ANN10 + T/P DeepONet10	**10k**	0.0444	0.0786	0.958	0.728	0.165	0.961	0.0649	0.92	0.99	0	18/18
5 ANN10 + T/P DeepONet10	**long**	0.224	0.116	3.09	0.778	0.338	5.48	0.291	0.916	0.996	0	18/18
6 T/P/ϕ DeepONet10	**10k**	0.0835	0.107	0.981	1	0.557	1	0.134	1.5	1.92	0	17/18
6 T/P/ϕ DeepONet10	**long**	0.215	0.0958	1.85	0.516	0.116	2.51	0.234	1.1	1.16	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.0672	0.121	1	0.561	0.21	1	0.138	0.998	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.101	0.0501	1.57	0.565	0.0648	2.52	0.164	0.994	1	0	18/18

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
Common ICs: 4, 5, 7, 32, 39, 44, 45, 57, 64, 68, 69, 73, 74, 76, 88.